
When Does Action-Type Supervision Help GUI Grounding?

Separating Real Effects from Evaluation Artifacts

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Abstract

GUI agents must predict an action type (*click, scroll, type, ...*) and a spatial target from a screenshot and an instruction. We study whether explicitly factoring these decisions helps a 2B-parameter agent, via a matched-compute comparison of five conditioning mechanisms on Qwen2-VL: a flat baseline (A), an auxiliary action-type loss (B), hard routing (C), an additive embedding hook (D-hook), and the prepended learned action embedding our original proposal advocated (D-token). Equally central, we report how our own first evaluation of this question misled us in three distinct ways, each of which we diagnose, quantify, and repair: (i) injecting conditioning via `inputs_embeds` silently bypasses Qwen2-VL’s M-RoPE position path and manufactured an 8σ *false negative*; (ii) train/validation splits taken as consecutive slices of one episode stream leaked shared task goals (24% of validation steps at the headline setting, up to 85% at scale); and (iii) a paired bootstrap that pooled (seed, example) predictions as independent units was anti-conservative, rejecting a true clustered null 21.2% of the time at nominal 5% (an episode-clustered replacement achieves 5.1%). After the statistical repair, the auxiliary-loss effect survives on the historical data (B–A $\text{hit@0.10} +0.051 [+0.020, +0.081]$, permutation $p = 0.0043$, versus the invalid pooled $p = 0.0002$); we then retrain all variants on frozen goal-disjoint splits and evaluate once on an untouched test set under a prespecified analysis. On that test set, B–A is [PENDING] (Holm $p =$ [PENDING]) and the end-to-end predicted-type pipeline versus A is [PENDING] (Holm $p =$ [PENDING]). The mechanism our proposal bet on is refuted as the winner: D-token is causally used at inference (wrong or zeroed embeddings severely hurt) yet never beats the plain auxiliary loss, and exploratory per-class decompositions locate the gain in click-vs-scroll/type disambiguation rather than field localization. We release the frozen splits, per-example predictions, and the clustered-inference code; we argue that offline step-level grounding studies at this scale are unusually easy to contaminate, and that the three artifacts we document are likely present in other evaluations.

1 Introduction

A GUI agent maps a screenshot and a natural-language instruction to a low-level action: a type (*click, scroll, type, drag, ...*) and, for spatial actions, a pixel-level target. State-of-the-art native agents (UI-TARS [13], OS-Atlas [11], SeeClick [4], ShowUI [7]) emit both in one autoregressive

stream, entangling the two decisions. A documented failure mode follows: commit to the wrong action type, then ground *coherently* within the wrong semantic frame – scrolling past a visible target instead of clicking it. This motivated our original hypothesis, formulated as an architecture bet: a learned action-type embedding, prepended to the instruction, should let a grounding model condition on *what* before deciding *where*, and should help most in low data.

This paper reports what happened when we tested that bet carefully, and it has two intertwined subjects. The first is the scientific question: *when does action-type supervision change GUI grounding, and through which mechanism?* The second is methodological: *which implementation and evaluation choices can create a false appearance of benefit or harm in studies of exactly this kind?* We came to the second subject involuntarily – three separate artifacts in our own pipeline each produced a wrong conclusion at some point during the project – and we believe the post-mortem is at least as useful to practitioners as the object-level result.

Concretely, we compare five matched-compute Stage-2 conditioning mechanisms on Qwen2-VL-2B [10] with LoRA, over AITW [9] with a Mind2Web [5] cross-dataset control, preceded by a Stage-1 action-type classifier study. Our contributions:

- **A controlled answer to the factorization question.** Auxiliary action-type supervision (B) improves grounding where action type is spatially predictive; the effect vanishes on a matched control where it is not. The proposal’s specific mechanism – a routable prepended embedding (D-token) – is causally used by the trained model yet never outperforms the far simpler auxiliary loss, refuting the architecture bet while supporting the broader thesis.
- **Three quantified evaluation pitfalls.** (1) A silent M-RoPE / `inputs_embeds` interaction that fabricated an 8σ false negative; (2) episode-stream train/validation leakage that is invisible at the episode level (2% of validation steps) but large at the shared-task-goal level (24%–85%); (3) pseudo-replicated significance testing whose anti-conservativeness we measure on synthetic clustered nulls (21.2% type-I error at nominal 5%).
- **A repaired, releasable protocol.** Pinned-revision, goal-disjoint frozen train/validation/test manifests with an untouched test region beyond every historical data access; episode-clustered paired inference with per-seed effects and validated calibration; and a prespecified confirmatory analysis committed before test access. All artifacts, per-example predictions, and analysis code are released.
- **An honest confirmatory result.** We report the prespecified untouched-test outcome for every primary contrast whatever its direction, alongside the exploratory history that motivated it.

2 Related Work

End-to-end GUI agents. The dominant paradigm treats action type, arguments, and coordinates as one autoregressive output. UI-TARS [13] exemplifies this with a unified action space across platforms and explicit thought traces. OS-Atlas [11] scales the same paradigm with 13M cross-platform GUI grounding instances and shows that action-label harmonization across web/Android/desktop is nontrivial. ShowUI [7] builds a compact VLA around Qwen2-VL with interleaved vision-language-action streaming. Our flat baseline (Variant A) replicates the joint-decoding paradigm of these systems.

Modular grounding. SeeClick [4] argues that grounding is the central bottleneck in visual GUI agents. UGround / SeeAct-V [6] separates planning from grounding at the text-vs-pixel boundary. GUI-Actor [14] replaces coordinate-token regression with an attention-based action head driven by a dedicated <ACTOR> token plus a verifier. Our work studies a different module boundary: between *action-type prediction* and *spatial grounding*, not between planning text and coordinates.

Reward decomposition. UI-R1 [8] applies RL fine-tuning with a rule-based reward that separates action-type accuracy from action-argument accuracy on AndroidControl. It is the closest training

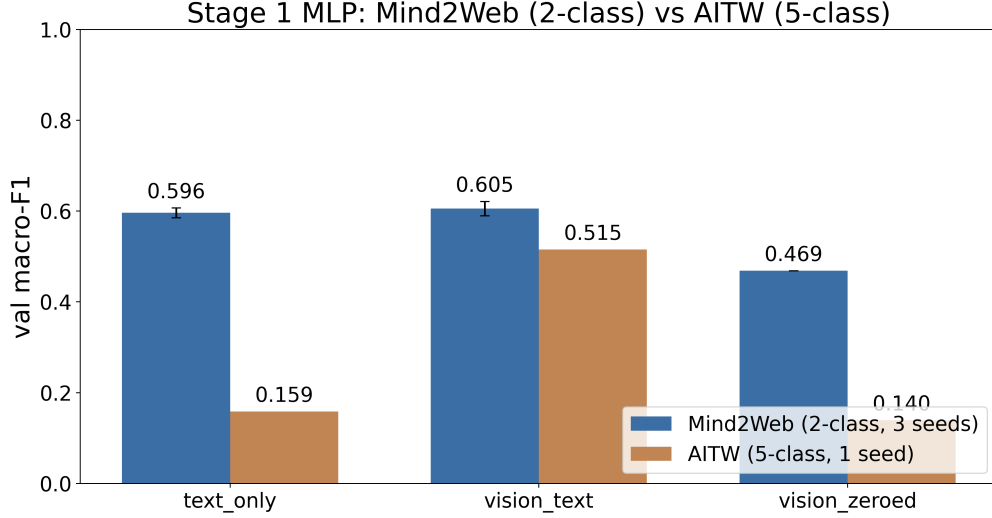


Figure 1: Stage 1 action-type classifier macro-F1 on Mind2Web (2 active classes) and AITW (5 active classes). Vision is essential on AITW (+0.414 over text-only) but redundant on Mind2Web (+0.009), because AITW instructions describe *goals* rather than physically specifying the action.

81 analog: type and location as separable optimization targets; we use supervised conditioning rather
 82 than RL. RegionFocus [2] refines grounding at inference; we change the training-time conditioning
 83 pathway. SayCan [1] and ReAct [12] provide the modular-decision precedent, and RT-2 [3] the
 84 unified token-level action generation paradigm we critically examine.

85 **Evaluation integrity.** Our methodological findings echo, in a GUI setting, long-standing concerns
 86 about test-set contamination and dependence-blind inference: grouped data require grouped splits
 87 and cluster-aware uncertainty, and repeated measurements of the same example must not be pooled
 88 as independent evidence. Mind2Web’s own metric design (element accuracy vs. operation F1 vs.
 89 step success) [5] reflects the type/location decomposition we study.

90 3 Data and Splits

91 3.1 Sources and preprocessing

92 We use two datasets under a canonical 8-class action taxonomy (*click*, *double_click*, *type*, *scroll*,
 93 *drag*, *hotkey*, *wait*, *finished*).

94 **AITW** [9] (via the `cjfcst/AITW_General` HuggingFace mirror, revision pinned to `5c0dc713`) is
 95 the primary Stage-2 benchmark: five active classes (click 55%, scroll 13%, finished 12%, type 11%,
 96 hotkey 9%). Screenshots in the mirror are raw RGB bytes without headers; we decode them against
 97 seven observed resolutions with a portrait-aspect fallback. DUAL_POINT events are split into taps
 98 (click) versus swipes (scroll) by normalized touch–lift distance (threshold 0.04). Two data mixes
 99 matter below: `all_with_coords` (tap, swipe, type – action type is spatially predictive because taps
 100 target small elements while swipes are gestures) and `taps_and_swipes` (a control where knowing
 101 the type carries little information about *where*).

102 **Mind2Web** [5] (multimodal release) provides the Stage-1 cross-dataset comparison and a Stage-2
 103 cross-dataset control; it is click/type-dominated with only two active classes at scale.

104 **Coordinates.** Targets are normalized to $[0, 1]^2$ and rendered as the string " (x, y) " on a $[0, 999]$
 105 grid; training loss is masked to the coordinate tokens.

106 3.2 Two evaluation protocols

107 **Exploratory protocol (Phases 1–7, historical).** Our original runs streamed the AITW training split
 108 in mirror order, filtered by data mix, and took *consecutive slices*: the first n_{train} steps for training
 109 and the next n_{val} for validation. This is the kind of split that looks innocuous in code review and is
 110 not: episodes and, worse, repeated task goals straddle the boundary (Section 5.2). Every historical
 111 number in this paper is labeled exploratory and is re-analyzed with dependence-aware statistics.

112 **Confirmatory protocol (Phase 9, frozen).** We rebuilt the data pipeline around a frozen manifest:
 113 the first 12,000 `all_with_coords` steps of the pinned mirror revision, grouped by *normalized task*
 114 *goal* (episodes never span goals), split goal-disjointly into train (9,511 steps), validation (1,660),
 115 and an **untouched test region** (829): test groups are drawn exclusively from beyond stream index
 116 5,500, past the furthest index any historical run ever loaded (5,250), so no test step was trained on,
 117 evaluated on, or inspected during the exploratory phase. Evaluation sets are frozen and identical
 118 for every variant and training size (250 validation / 600 test steps for `all_with_coords`; 200/400
 119 for the control mix); training subsets are nested prefixes. The manifest, split hashes, 17 disclosed
 120 near-duplicate goal pairs (token Jaccard ≥ 0.8), duplicate checks, and the prespecified analysis
 121 (contrasts, metrics, corrections, failed-run rules) were committed to version control *before* any test-
 122 set evaluation ran. We claim “decisions frozen before test access,” documented by commit history –
 123 not third-party preregistration.

124 4 Methods

125 4.1 Stage 1: action-type classifier

126 A *frozen* Qwen2-VL-2B encodes screenshot+instruction; mean-pooled final hidden states (1536-
 127 d) feed a 3-layer MLP trained with class-weighted cross-entropy (AdamW, lr 10^{-4} , batch 128, 8
 128 epochs, best epoch by validation macro-F1). Modes: `text_only` vs. `vision_text`.

129 4.2 Stage 2: five conditioning mechanisms

130 Stage 2 fine-tunes Qwen2-VL-2B with LoRA (rank 16 on q/k/v/o projections, $\approx 0.2\%$ of parameters)
 131 to emit the coordinate string; all variants share data, optimizer (lr 2×10^{-5} , 2 epochs, batch 1), and
 132 evaluation. The only difference is how action-type information enters:

133 **A (flat baseline).** No conditioning; replicates joint decoding.

134 **B (auxiliary loss).** A linear action-type head on the pooled representation adds cross-entropy to the
 135 grounding loss ($\lambda=1$); *no action information at inference*.

136 **C (hard routing).** Trained to emit “{word} (x, y)”; at inference the gold action word is forced
 137 as the decode prefix.

138 **D-hook (additive embedding).** A learned table $E \in \mathbb{R}^{8 \times d}$; the row for the current action type is
 139 added to *all* token embeddings via a forward hook, preserving the normal `input_ids` path (embed-
 140 ding norm at convergence ≈ 0.07 at the headline setting).

141 **D-token (prepended token – the original hypothesis).** A sentinel `<|action_slot|>` token is
 142 prepended to the instruction and its embedding replaced in place by $E[a]$, M-RoPE-correctly (the
 143 token is in `input_ids`); norm at convergence ≈ 2.2 , a strong routable signal.

144 **Objective.** With screenshot s , instruction g , action type a , coordinate tokens $y_{1:T}$ and variant-
 145 specific conditioning c :

$$\mathcal{L}_{\text{coord}} = - \sum_{t=1}^T \log p_{\theta}(y_t \mid y_{<t}, s, g, c), \quad (1)$$

146 with $c = \emptyset$ for A and B, $c = E[a]$ for D-hook/D-token, and the forced word for C; B adds
 147 $\lambda \text{CE}(h_{\phi}(\tilde{z}), a)$ (Eq. 1 otherwise unchanged).

historical cell	val steps	sharing a train episode	sharing a train goal
all_wc, $n=300$	250	4 (2%)	14 (6%)
all_wc, $n=500$	250	4 (2%)	47 (19%)
all_wc, $n=800$	250	5 (2%)	49 (20%)
all_wc, $n=1200$	250	4 (2%)	61 (24%)
all_wc, $n=2500$	250	10 (4%)	133 (53%)
all_wc, $n=5000$	250	2 (1%)	213 (85%)
taps_swipes, $n=1000$	200	0 (0%)	48 (24%)

Table 1: Leakage in the historical (exploratory) splits: fraction of validation steps sharing an episode or a normalized task goal with that cell’s training slice. Goal sharing, not episode straddling, is the dominant channel.

End-to-end pipeline. The deployable configuration trains D-hook, then a small MLP head on frozen-backbone features (Stage 1 in-job), and evaluates grounding under *predicted* types; gold-type results are oracle ceilings, never deployable claims.

4.3 Evaluation metrics

hit@ r : fraction of predictions within normalized L2 distance r of the target ($r \in \{0.05, 0.10, 0.25\}$; headline 0.10). **Mean L2:** mean normalized distance (parse failures scored at the $\sqrt{2}$ sentinel and counted as misses; parse rates reported). Statistical methodology is Section 5.3.

5 Three Evaluation Pitfalls, Quantified

5.1 Pitfall 1: M-RoPE vs. inputs_embeds

Qwen2-VL assigns 3D (temporal, height, width) position IDs to visual tokens via an `input_ids`-keyed path. Our first D-token implementation injected the action embedding by passing `inputs_embeds` directly, which silently bypasses that path: image-token positions were computed without the action slot in the sequence, misaligning multimodal position embeddings for *every* visual token. The observed effect was a confident 8σ “A beats D” result – a pure artifact. Rebuilding the same mechanism through the `input_ids` path (D-hook, D-token) erased the deficit. Any conditioning study on M-RoPE-family models that touches `inputs_embeds` should verify position-ID computation explicitly; the artifact is silent, large, and directionally misleading.

5.2 Pitfall 2: stream-slice splits leak task goals

Table 1 quantifies the historical protocol’s leakage by reconstructing, for every historical cell, which episodes and normalized task goals each validation step shared with that cell’s training slice. Episode-level overlap – the leakage one usually checks – is small (2% at the headline cell). The dominant channel is *shared task goals*: 24% of headline validation steps have a goal string that also appears in training, rising to 85% at $n_{\text{train}}=5000$ – AITW goals repeat across many episodes, so consecutive-slice splitting almost guarantees template overlap at scale. Val-set difficulty therefore varies with n_{train} (each size had a different val slice), quietly confounding the historical scaling curve. The confirmatory protocol removes both channels by construction (goal-disjoint splits; Section 3.2).

5.3 Pitfall 3: pseudo-replicated significance

Our original headline significance pooled per-example distances across seeds – treating 3×250 (seed, example) rows as 750 independent paired observations. Predictions of the same example by different seeds are strongly correlated, and examples within an episode share state; the pooled bootstrap therefore overstates evidence. On synthetic nulls matching the data’s dependence structure (episode and example effects shared across seeds; zero true difference), the pooled test rejects 21.2% of the time at nominal 5% (1000 simulations), while the replacement below achieves 5.1%.

Variant	Mind2Web macro-F1	AITW macro-F1
text_only	0.597 $\pm$ 0.011	0.141 $\pm$ 0.032
vision_text	0.605 $\pm$ 0.016	0.555 $\pm$ 0.036
vision_zeroed	0.469 $\pm$ 0.000	0.110 $\pm$ 0.051
Vision delta	+0.009	+0.414
TF-IDF + LogReg	0.622	0.168
Majority floor	0.472	0.140

Table 2: Stage 1 action-type classification (3 seeds, mean $\pm$ std; exploratory splits). On Mind2Web the instruction text alone contains the action type; on AITW it does not, and vision contributes +0.414 macro-F1. `vision_zeroed` collapsing to the majority floor is the sanity check that features, not priors, drive the result.

variant	hit@0.10	mean L2 $\downarrow$	Δ hit@0.10 vs. A [95% CI]	perm. p
A (flat)	0.255 $\pm$ 0.021	0.392	—	—
B (aux loss)	0.305 $\pm$ 0.012	0.362	+0.051 [+0.020, +0.081]	0.0043
C (hard routing)	0.279 $\pm$ 0.032	0.412	+0.024 [-0.016, +0.063]	0.278
D-hook (additive)	0.300 $\pm$ 0.028	0.365	+0.045 [+0.011, +0.085]	0.0227
D-token (prepending)	0.269 $\pm$ 0.025	0.375	+0.015 [-0.018, +0.047]	0.437

Table 3: Exploratory main ablation (AITW `all_with_coords`, $n_{\text{train}}=1200$, 3 seeds, historical leaky splits). Episode-clustered Δ vs. A with 95% cluster-bootstrap CIs and sign-flip permutation p . Under the repaired analysis, B survives (Holm-adjusted $p = 0.017$ across the four vs.-A contrasts); D-hook is marginal after correction (Holm $p = 0.068$); C and D-token are not significant.

181 **Repaired inference.** For each contrast we (i) average each example’s paired difference over seeds,
182 (ii) cluster examples by episode, (iii) bootstrap clusters (10,000 resamples) for 95% CIs, and (iv) use
183 an episode-level sign-flip permutation test (10,000 permutations; p -values reported no smaller than
184 Monte Carlo resolution – never $p=0$). **Estimand.** This tests the *fixed trained checkpoints* on new
185 data from the eval population. It is *not* calibrated for claims about the training procedure: adding
186 run-level (seed) offsets to the synthetic null drives the episode-clustered test to 55% rejection (and
187 the pooled test to 70%). With only three seeds per variant we therefore report per-seed means every-
188 where and make no procedure-level significance claims. The retracted pooled p -values (including a
189 headline $p=0.0002$) should not be cited.

190 6 Exploratory Results (historical splits, repaired statistics)

191 All numbers in this section come from the leaky historical splits and are therefore *exploratory*;
192 they are shown with episode-clustered statistics, and Section 7 gives the prespecified untouched-test
193 versions of every primary claim.

194 6.1 Stage 1: vision is essential exactly where instructions under-specify the action

195 Table 2 and Figure 1: on Mind2Web, instructions name the action (“click the submit button”), so
196 vision adds +0.009 macro-F1 over text; on AITW, instructions state goals (“open a new Chrome
197 private window”), text-only collapses toward the majority class, and vision adds +0.414. A TF-IDF
198 ceiling of 0.168 on AITW confirms text is uninformative there regardless of classifier strength. This
199 asymmetry decides where Stage-2 conditioning could matter at all.

200 6.2 Main ablation under clustered statistics

201 Table 3 re-scores the historical headline with the repaired statistics (per-seed values in Figure 3).
202 Three observations. First, the auxiliary-loss effect survives: B–A on hit@0.10 is +0.051 [+0.020,

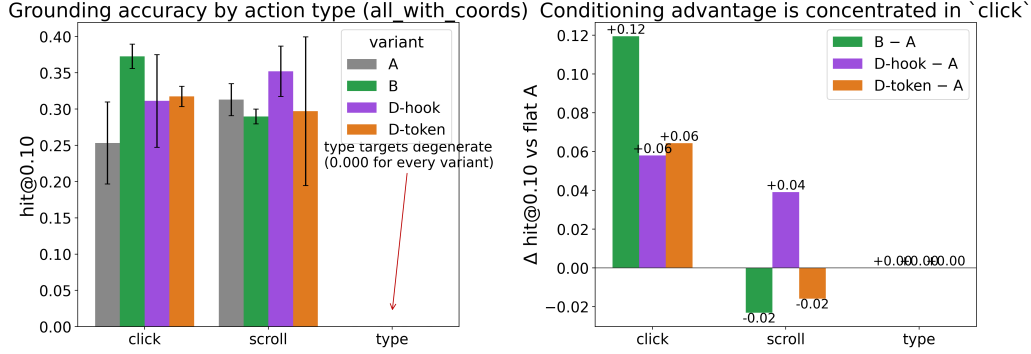


Figure 2: Exploratory hit@0.10 by action class (all_with_coords). The conditioning advantage concentrates in *click*; *type* is degenerate on AITW (all variants score exactly 0.000, mean L2 at the $\approx \sqrt{2}$ corner-to-corner distance).

+0.081] ($p = 0.0043$), where the invalid pooled analysis had claimed $p = 0.0002$ – the conclusion stands but the evidence was overstated by an order of magnitude. Second, D-hook weakens to marginal after multiplicity correction. Third, the comparison our proposal cared about most – D-token vs. B – is *no longer significant* (-0.036 [$-0.077, +0.005$], $p = 0.114$): on the historical data we cannot even claim the prepended embedding is significantly *worse* than the auxiliary loss, only that it shows no sign of being better.

6.3 Per-class decomposition: the gain is click disambiguation

Breaking hit@0.10 down by class (Figure 2): B gains +0.120 on click (68% of examples) with a small scroll cost; C has the largest click gain (+0.122) but destroys scroll (-0.144) by forcing a de-code prefix that derails the gesture class; D-hook alone improves both (click +0.058, scroll +0.039). The *type* class scores exactly 0.000 across all variants with mean normalized L2 at or slightly above $\sqrt{2}$, the corner-to-corner distance (a few predictions land outside the unit square): AITW *type* events carry degenerate touch coordinates and are ungroundable by any model. This corrects our earlier “*type*→text-field localization” interpretation: the real signal is **click-vs-scroll/type disambiguation** for the precise-target class, and the “*type*” rows merely add a constant drag that cancels in deltas.

6.4 Control conditions

On *taps_and_swipes* (action type spatially uninformative), the historical 3-seed means show no conditioned variant meaningfully above A (A 0.390 ± 0.008 , D-hook 0.392 ± 0.035 – a difference well inside one seed-std – with B 0.368 ± 0.009 and C 0.325 ± 0.032 at or below A on hit@0.10); per-example logs were not retained for A on this mix, so the clustered version of this control appears only in the confirmatory rerun (Table 6). On Multimodal-Mind2Web (click/type-dominated), D-hook vs. A was -0.006 hit@0.10 / $+0.018$ hit@0.25 over 2 seeds – a null, retained here deliberately: conditioning does not help where action type carries no spatial information, on either dataset.

6.5 Low-data behavior: no clean curve

Our proposal predicted the advantage is a low-data prior. The historical grid ($n \in \{300, 500, 800\}$, 3 seeds) does *not* show a clean monotone: B–A on hit@0.10 is +0.067 at $n=300$ but +0.011 at $n=500$ and -0.040 at $n=800$; D-hook is more stable (+0.069, +0.001, +0.048) and leads hit@0.25 and mean L2 at every low-data size. The two single-seed high-end cells disagree with each other:

contrast	Δ hit@0.10 [95% CI]	Δ hit@0.25 [95% CI]	perm. p (hit@0.10)
wrong – gold	−0.192 [−0.233, −0.149]	−0.329 [−0.372, −0.284]	$\leq 9.999e - 05$
zero – gold	−0.093 [−0.128, −0.059]	−0.112 [−0.145, −0.079]	$\leq 9.999e - 05$

Table 4: Exploratory D-token sensitivity (episode-clustered, 3 seeds): evaluating the same trained model with wrong or zeroed action embeddings severely degrades grounding. This establishes the embedding is *used* at inference – a sensitivity/oracle analysis, not evidence that conditioning caused any gain over A.

PENDING: confirmatory rerun results not yet rendered

Table 5: **Confirmatory** main ablation on the untouched, goal-disjoint AITW test set ($n=600$ steps; $n_{\text{train}}=1200$; 3 seeds; mean $\pm$ std across seeds). Δ vs. A: episode-clustered 95% CI and permutation p (Holm-adjusted across the four primary contrasts where applicable).

at $n=5000$ the deltas shrink and partially reverse (B–A hit@0.25 goes negative), while at $n=2500$ D-hook posts the *largest* conditioning deltas anywhere in the study (+0.076 hit@0.10, +0.204 hit@0.25) – single-seed noise on a differently-leaky val slice, cutting in the direction *favorable* to conditioning. We report all of this as: *inconclusive on the low-data-prior hypothesis – noisy, non-monotone, and confounded by per-size val sets* (Figure 4); the prespecified low-data claim is evaluated on the frozen test set in Section 7. An earlier draft of this work called the low-data prediction “confirmed”; that claim did not survive this audit.

6.6 The e2e pipeline claim also weakens

The historical end-to-end result (predicted-type pipeline 0.271 vs. A 0.255 hit@0.10) was previously reported as a win. Under clustered statistics the paired difference is +0.016 [−0.017, +0.052] ($p = 0.424$) – not significant; the oracle-type version is marginal (+0.037 [+0.005, +0.076]). The deployable-pipeline question is therefore decided by the confirmatory run (P4, below), not the historical data.

6.7 D-token causal-use sensitivity

Feeding the trained D-token model a *wrong* action id collapses grounding, and zeroing the embedding hurts substantially (Table 4). The refutation of the proposal’s mechanism is therefore precise: the prepended embedding is causally used, it is simply not the best way to exploit action-type supervision. Figure 5 shows one example where correct conditioning concentrates more attention mass on the screenshot; across the three examples we probed the pattern is not uniform, so we present it strictly as illustration, not evidence.

7 Confirmatory Results (untouched test, prespecified)

All variants were retrained from scratch on the frozen goal-disjoint splits (identical hyperparameters, seeds 42–44) and evaluated once on the untouched 600-step test set; the four primary contrasts, the headline metric (hit@0.10), the clustered test, and Holm correction were committed before any test evaluation ran (Section 3.2).

P1 (auxiliary supervision). B vs. A on hit@0.10: [PENDING], permutation $p =$ [PENDING], Holm $p =$ [PENDING]; mean-L2 delta [PENDING] ($p =$ [PENDING]).

PENDING: confirmatory rerun results not yet rendered

Table 6: Confirmatory control mix (taps_and_swipes) on its frozen 400-step test set: when action type is not spatially predictive, conditioning should not help; this is the falsifiable half of the thesis.

PENDING: confirmatory rerun results not yet rendered

Table 7: Confirmatory low-data grid on the *same* frozen test set for every training size (nested training prefixes), removing the per-size-val-set confound of Figure 4.

259 **P2 (inference-time conditioning).** D-hook vs. A: [PENDING] (Holm p = [PENDING]).
260 **P3 (the proposal’s mechanism).** D-token vs. B: [PENDING] (Holm p = [PENDING]).
261 **P4 (deployable pipeline).** End-to-end predicted-type pipeline vs. A: [PENDING] (Holm p =
262 [PENDING]); Stage-1 test accuracy [PENDING], oracle ceiling [PENDING] vs. predicted
263 [PENDING] (paired gap [PENDING]).
264 Secondary contrasts, per-class decompositions, parse rates, and per-seed values for every run are in
265 the released `results/phase9_rerun/rerun_analysis.json`; per-seed means for each primary
266 contrast appear alongside the released analysis, and no procedure-level significance claim is made
267 from three seeds (Section 5.3).

268 8 Limitations

269 Offline step-level hit@radius is not long-horizon task success: it scores logged single steps, without
270 interaction, recovery, or compounding-error dynamics – the motivating failure mode is measured
271 here only via its step-level signature. The backbone is a single 2B model; conclusions are about
272 this scale and training recipe, not about GUI agents generally, and our baselines are small matched-
273 compute models trained by us, not state-of-the-art systems. The confirmatory estimand is fixed
274 checkpoints, not the training procedure (three seeds; per-seed effects reported). AITW’s type coordi-
275 nates are degenerate, so “grounding” on `all_with_coords` is effectively click/scroll grounding
276 plus a constant; the click-concentration analysis is post-hoc on exploratory data and only its prespec-
277 ified aggregate versions are confirmatory. Goal strings are our dependence proxy; 17 near-duplicate
278 train/test goal pairs (disclosed in the manifest) survive normalization, and subtler sharing (same app,
279 similar screens) is not modeled. The Mind2Web control used 2 seeds without per-example logs. Fi-
280 nally, the mirror we pin is a third-party repackaging of AITW; we verified revision stability but not
281 byte-level provenance against the original release.

282 9 Conclusion

283 The broad factorization thesis survives careful evaluation in narrowed form: auxiliary action-type
284 supervision helps a small GUI grounding model where action type is spatially predictive, the effect
285 concentrates in click disambiguation, and it disappears on matched controls – with the confirmatory
286 test-set verdicts for all four prespecified contrasts reported above exactly as they came out. The
287 specific architecture our proposal advocated – a learned prepended action embedding – is refuted
288 as the winning mechanism despite being demonstrably used by the model. En route we hit, diag-
289 nosed, and repaired three evaluation artifacts (a silent M-RoPE position bug producing an 8σ false
290 negative; goal-level split leakage reaching 85%; pseudo-replicated statistics with 21.2% measured
291 type-I error), each individually capable of flipping the paper’s conclusion. We suspect none of them
292 is unique to us: stream-sliced splits, embedding-injection shortcuts, and pooled per-seed bootstraps
293 are all common patterns. The frozen splits, per-example predictions, clustered-inference code, and
294 prespecification are released so both the object-level result and the audit are reproducible.

295 Contributions

296 **A. Chauhan:** training and evaluation infrastructure, Stage-2 conditioning variants, cloud experi-
297 ment orchestration, statistical reanalysis and confirmatory protocol, writing. **A. Ilyasov:** Stage-1

298 classifier study, data pipeline and error analysis, figures, editing. Both authors contributed to prob-
299 lem formulation and experimental design and approved the final manuscript.

300 **Generative AI Usage Statement**

301 Large-language-model assistants (Anthropic Claude) were used extensively under the authors’ direc-
302 tion: writing and debugging training, evaluation, and statistical-analysis code; orchestrating cloud
303 experiments; drafting the integrity audit that led to Sections 5 and 7; and drafting and editing this
304 manuscript. The authors specified the research questions and methodology, reviewed the analyses,
305 verified reported numbers against the committed result artifacts (every table is generated program-
306 matically from those artifacts), and take full responsibility for all content.

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335 **A Additional exploratory figures**

336 These figures accompany Section 6 (exploratory results on the historical splits) and are referenced
337 from the main text.

Stage 2 ablation (AITW all_with_coords): $B \approx D\text{-hook} > C > A$

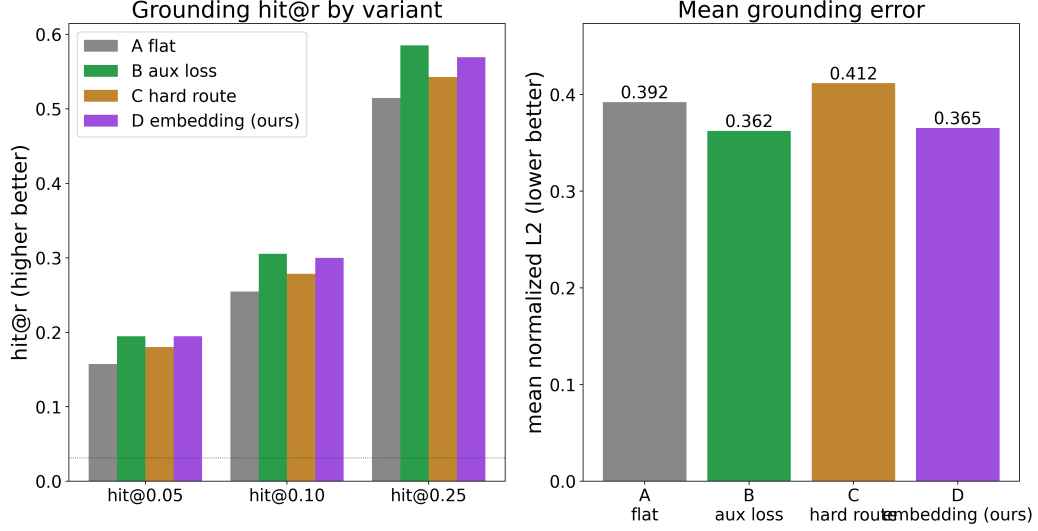


Figure 3: Exploratory ablation on all_with_coords (historical splits; per-seed values). Shown for continuity with the historical record; Table 3 carries the calibrated statistics and Table 5 the confirmatory test-set versions.

Data-scaling: action-type conditioning advantage vs training size
(AITW all_with_coords; 300-1200 are 3-seed means, 2500/5000 single seed)

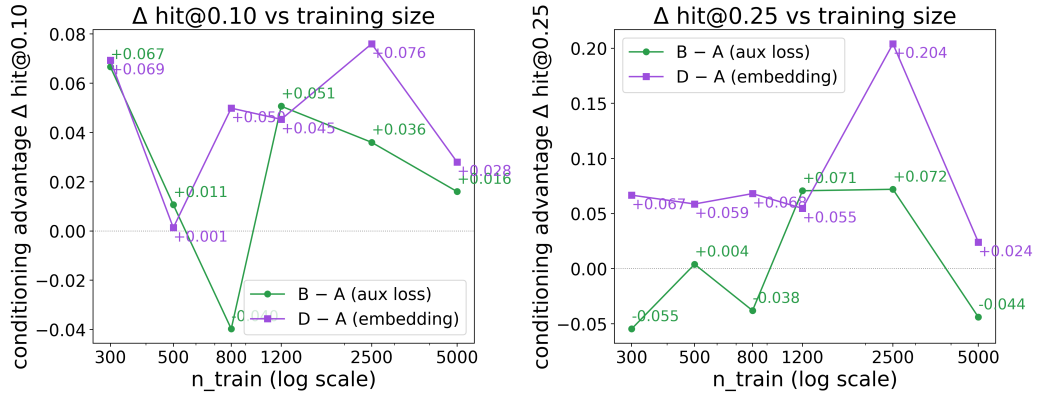


Figure 4: Exploratory conditioning advantage ($\Delta \text{hit@0.10}$ vs. A) across n_{train} . Caveat from Section 5.2: each historical size used a *different* val slice with different goal leakage (up to 85%), so this curve is directional at best; the confirmatory grid (Table 7) fixes one frozen test set for all sizes.



Figure 5: Image-region attention for a held-out *click* example under three action-type conditionings (qualitative illustration; screenshot attention mass 0.204 under *click* vs. 0.132/0.110 under *type/scroll*).